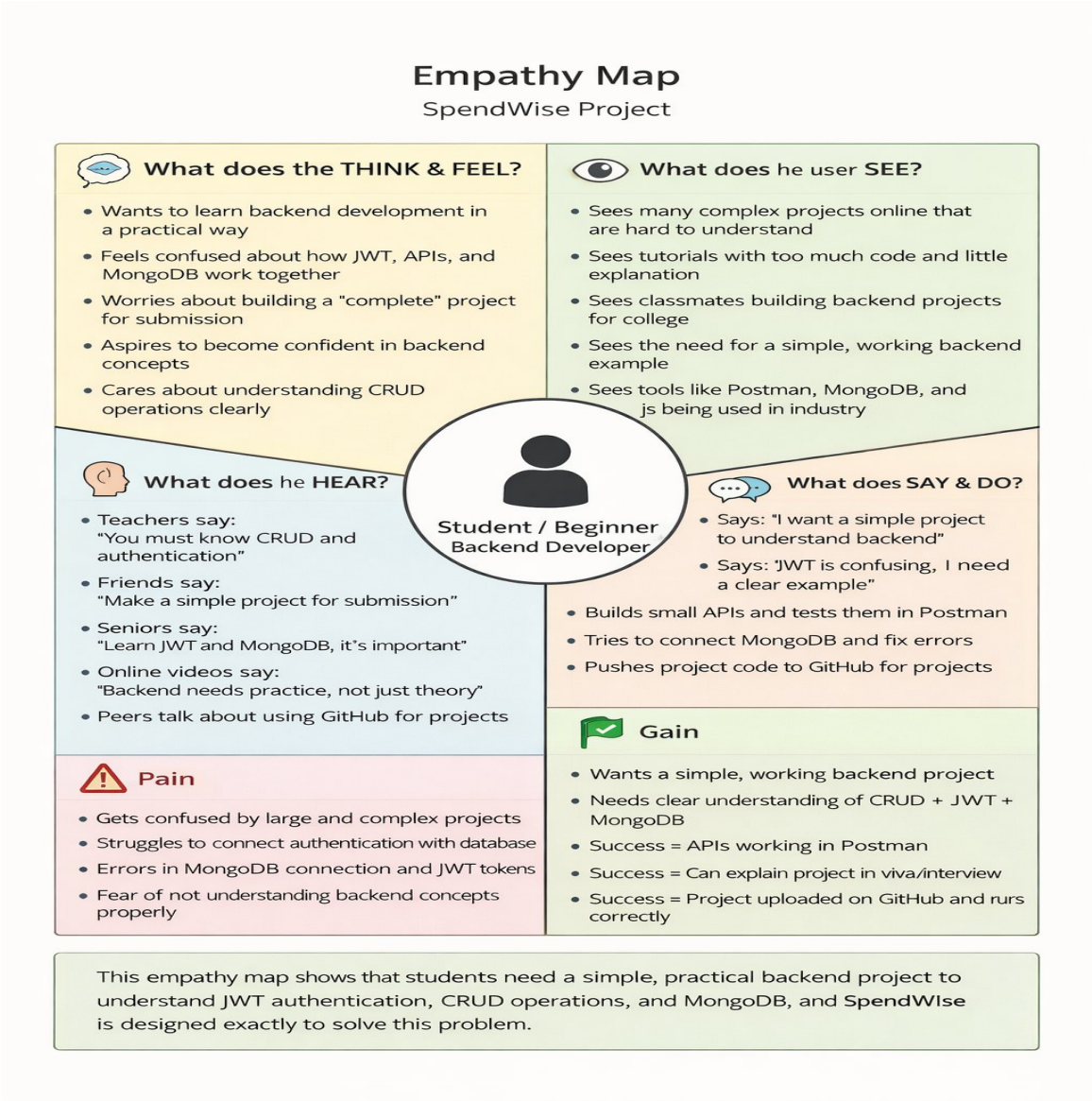
SpendWise: A Simple Backend Learning Project
Ideation Phase
Empathize & Discover

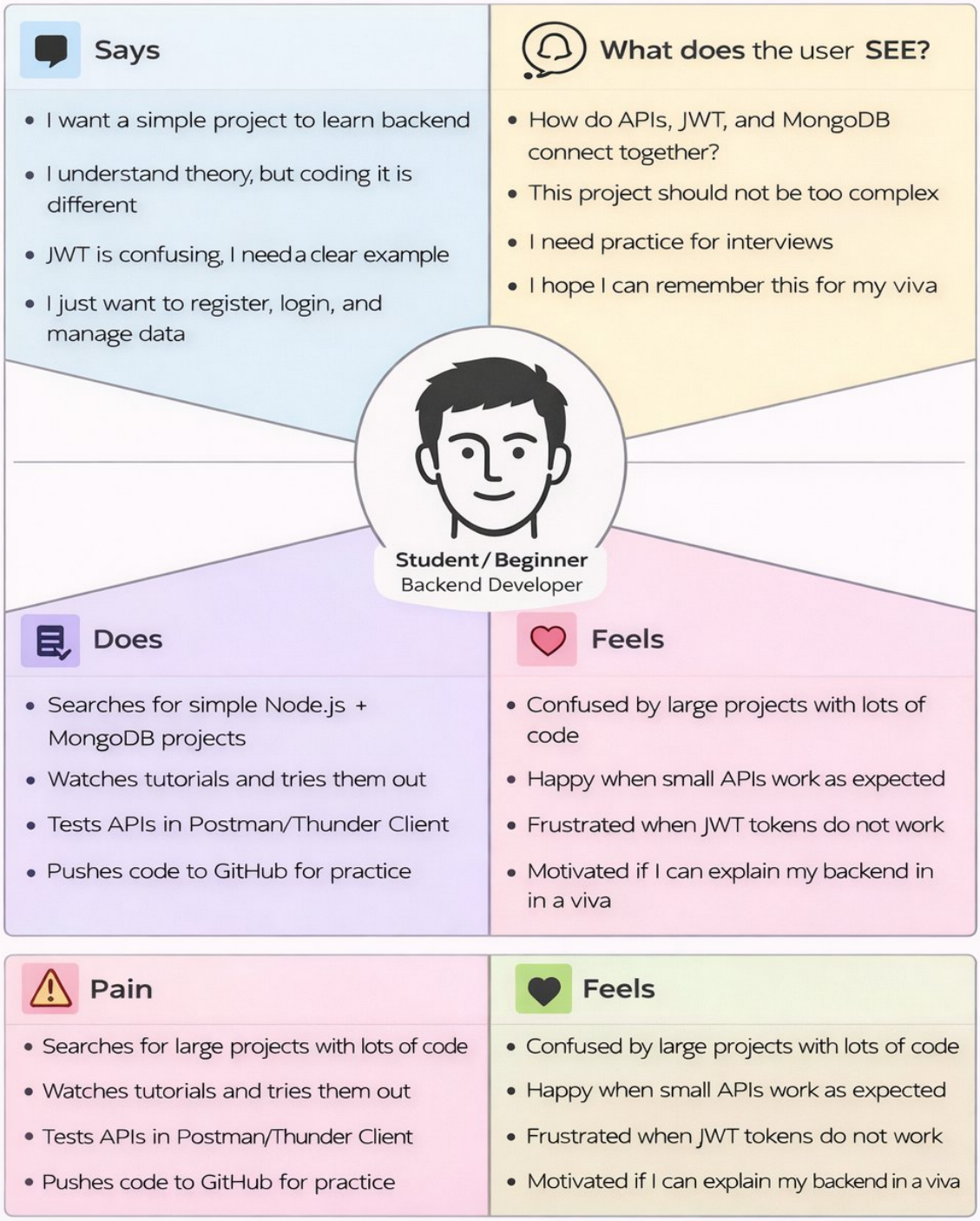
Date	6 march 2026
Team ID	SWTID-2026-9594
Project Name	SpendWise
Maximum Marks	4 Marks

Empathy Map Canvas:

An empathy map is a simple way to understand a beginner's thoughts, feelings, and actions while learning backend development. It helps identify how beginners work with APIs, JWT authentication, and MongoDB, and what problems they face with complex code. Creating this empathy map allows the project to be viewed from the beginner's perspective, ensuring the solution stays simple, practical, and easy to understand.

Example:





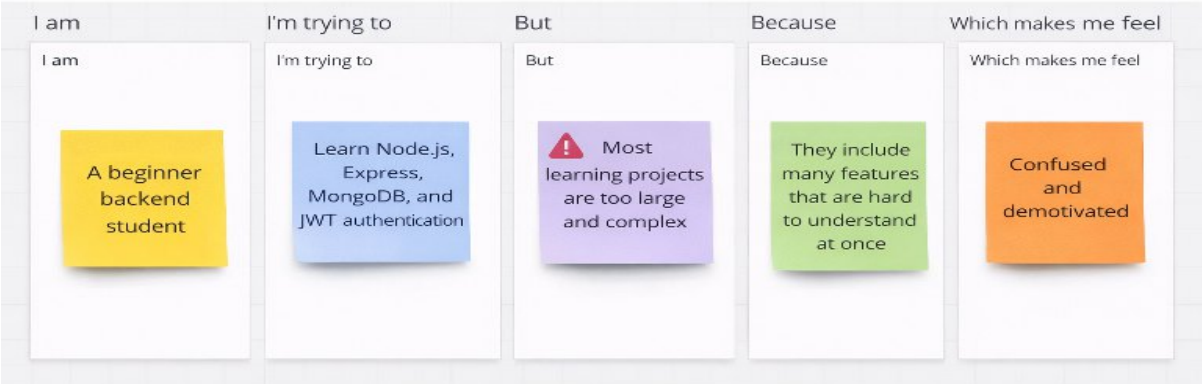
Customer Problem Statement Template:

A customer problem statement helps understand the user's point of view while learning from

a product. In the SpendWise project, it focuses on the needs of beginner backend students who want to learn Node.js, Express, MongoDB, and JWT in a simple way. It helps identify problems like confusion with complex projects and difficulty connecting concepts, so the solution stays clear, useful, and easy to learn.

I am	A beginner backend student
I'm trying to	Learn Node.js, Express, MongoDB, and JWT authentication
but	Most learning projects are too large and complex
because	They include many features that are hard to understand at once
which makes me feel	Confused and demotivated
	Confused and demotivated

example:



Problem	I am	I'm trying to	But	Because	Which makes

Statement (PS)	(Customer)				me feel
PS-1	A beginner backend student	Learn Node.js, Express, MongoDB, and JWT authentication	Most learning projects are too large and complex	They include many features that are hard to understand at once	Confused and demotivated
PS-2	A college student	Build a simple backend project for practice and submission	I don't find a clear example focused only on CRUD and authentication	Tutorials often jump to advanced topics without explaining basics	Unsure about my understanding
PS-3	A beginner developer	Connect JWT authentication with MongoDB and APIs	I keep getting errors in tokens or database connection	The flow between login, token, and protected routes is not clear	Frustrated and stuck
PS-4	A student preparing for viva or interviews	Explain CRUD operations and JWT authentication with a project	I don't have a simple end-to-end backend example	Most available projects are either too basic or too advanced	Nervous and underprepared

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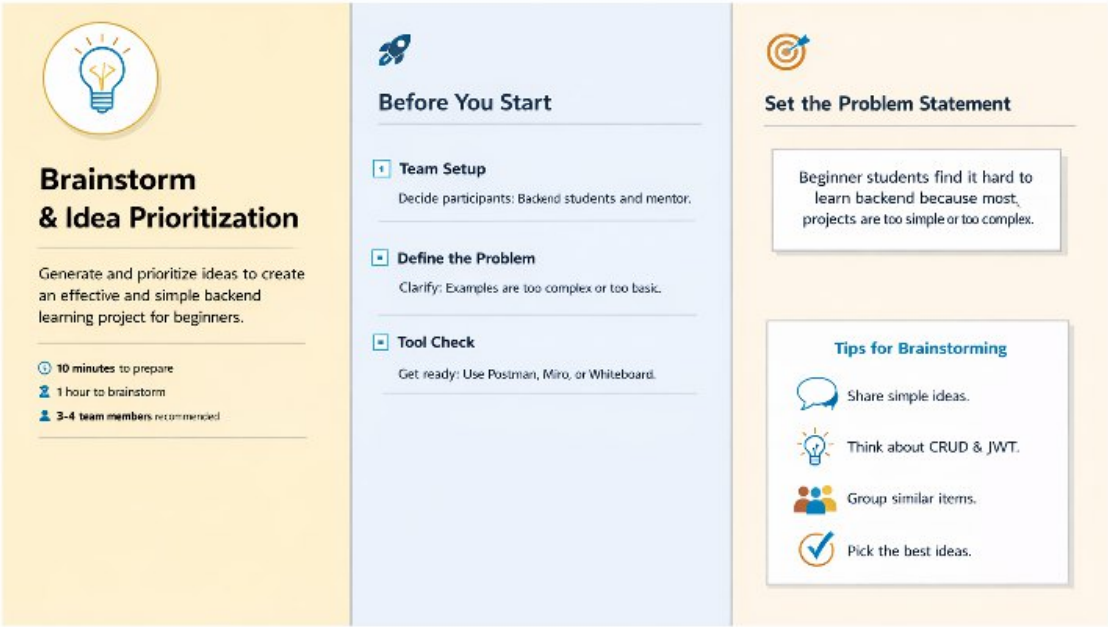
Brainstorm & Idea Prioritization Template:

Brainstorming creates an open environment where team members can freely share ideas to solve a problem. For SpendWise, the goal is to generate simple, practical ideas that help students learn backend development. All ideas are first listed without judging, then grouped, and finally prioritized based on usefulness and feasibility for beginners.

Step-1: Team Gathering, Collaboration and Select the Problem Statement



2

Brainstorm

Write down any ideas that come to mind to support the backend learning project.

10 minutes

TIP

Think about what you struggled to understand. Share ideas that keep the project simple, focused, and easy to learn.

Person 1	Person 2	Person 3	Person 4
Implement Register & Login with JWT authentication.	CRUD API for adding, viewing, updating, and deleting user expenses.	Use MongoDB with Mongoose to store users and expenses.	Simple and clear folder structure with routes, controllers, and models.

3

Group ideas

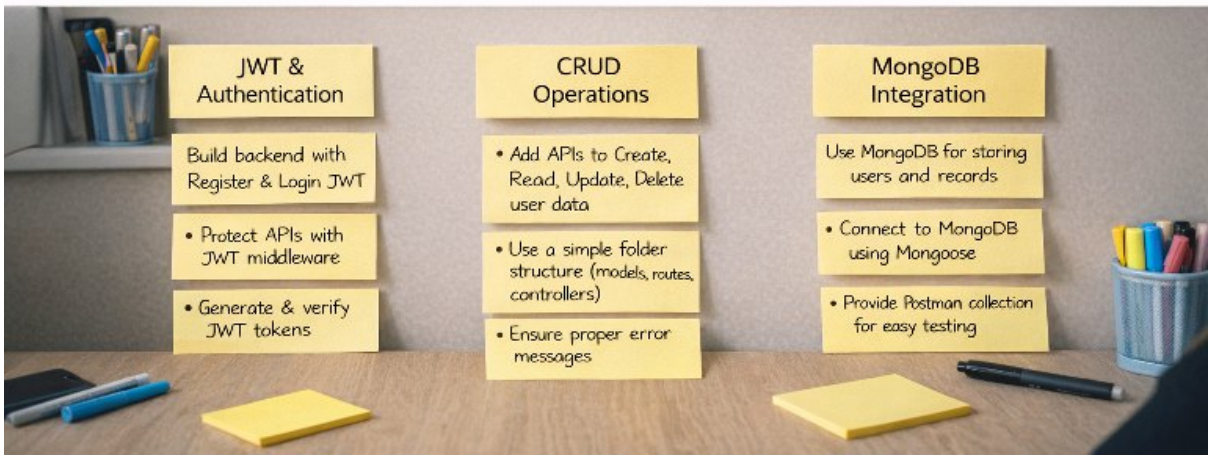
Take turns sharing youideas and group similar sticky notes into useful themes for **SpendWise**. Give each group a clear label describing the theme once all sticky notes are clustered.

20 minutes

TIP

Add tags to sticky notes to find, organize, and categorize ideas.

- Key tags to consider:
- CRUD Operations
- JWT & Authentication
- MongoDB Integration

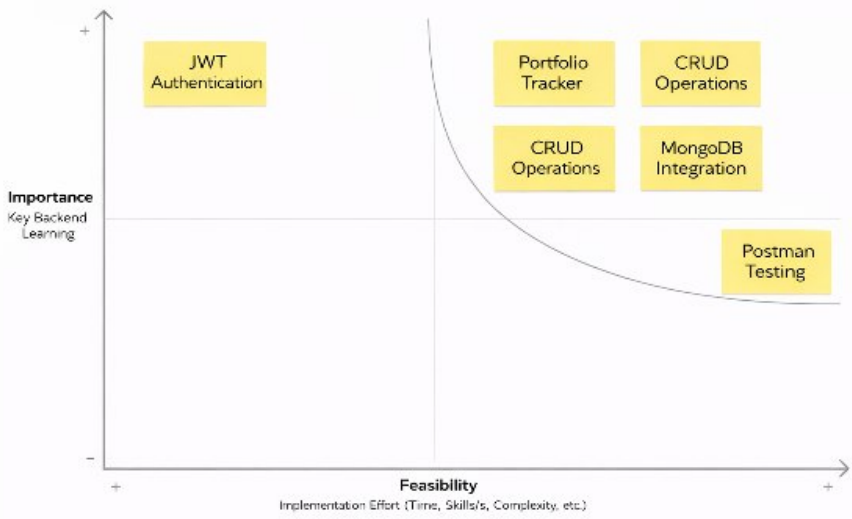


4

Prioritize

Align as a team to prioritize **SpendWise** ideas based on their importance and feasibility. Place sticky notes on the grid to determine which ideas are important and which are feasible.

20 minutes

TIP

Place ideas that are critical to learn backend in the top-right section (high importance and feasibility). Ideas that are nice-to-have but not crucial can go in the bottom-right section. Any less important or feasible ideas can go in the...